MAC TEAM PLAN

Final Project: PySpark + MongoDB + SQL Server + Power BI

Team Members: Mac User 1 & Mac User 2

Objective

The Mac team is responsible for:

1. Data Ingestion
2. MongoDB Atlas Setup
3. PySpark Processing & Analysis

The goal is to prepare clean and analyzed data that will later be consumed by the Windows team for SQL Server and Power BI visualization.

Team Structure

Member	Responsibility
Mac User 1	MongoDB & Data Ingestion
Mac User 2	PySpark Analysis & Data Processing

Architecture

Raw Dataset (CSV)

↓

MongoDB Atlas

↓

PySpark

↓

Processed Results

↓

SQL Server (Windows Team)

↓

Power BI Dashboard (Windows Team)

Mac User 1 Tasks

MongoDB & Data Ingestion

Goal

Load the dataset into MongoDB Atlas and verify successful ingestion.

Step 1: Create MongoDB Atlas Cluster

1. Create MongoDB Atlas account
2. Create M0 Free Cluster
3. Create database user
4. Whitelist team IP addresses
5. Obtain connection string

Example:

mongodb+srv://username:password@cluster.mongodb.net/

Step 2: Prepare Dataset

Tasks:

- Inspect dataset columns
- Remove duplicate rows
- Check missing values
- Save cleaned CSV

Output:

data/raw_dataset.csv

Step 3: Load Data into MongoDB

Create:

ingestion/load_mongodb.py

Responsibilities:

- Read CSV
- Convert records to JSON
- Insert into MongoDB collection

Verify:

- Record count
- Sample documents
- Collection creation

Deliverables

- MongoDB Atlas Cluster
- MongoDB Collection
- load_mongodb.py
- Screenshots of database

Mac User 2 Tasks

PySpark Analysis

Goal

Connect PySpark to MongoDB and perform analysis.

Step 1: Configure PySpark

Install:

pip install pyspark

Verify:

from pyspark.sql import SparkSession

Step 2: Connect PySpark to MongoDB

Create:

spark/connect_mongodb.py

Read collection:

df = spark.read.format("mongo").load()

Verify:

df.show()

df.printSchema()

Step 3: Exploratory Data Analysis

Perform:

Analysis 1

Summary Statistics

df.describe()

Analysis 2

Aggregation

Examples:

- Average value by category
- Count by status
- Total amount by group

Analysis 3

Correlation Analysis

Example:

df.stat.corr("column_a", "column_b")

Step 4 (Optional Bonus)

Build a simple ML model using:

- Logistic Regression
- Random Forest

Using:

pyspark.ml

Step 5

Export Processed Results

Save:

processed_results.csv

and/or

processed_results collection

Deliverables

- connect_mongodb.py
- analysis.py
- processed_results.csv
- Analysis screenshots

Shared Folder Structure

final-project/

├── data/ │ └── raw_dataset.csv

├── ingestion/ │ └── load_mongodb.py

├── spark/ │ ├── connect_mongodb.py │ └── analysis.py

├── outputs/ │ └── processed_results.csv

├── screenshots/

└── README.md

Timeline

June 11-13

Mac User 1

- Setup MongoDB Atlas
- Load Dataset
- Verify Data

Mac User 2

- Setup PySpark
- Test MongoDB Connection

June 14-15

Mac User 2

- Perform Analysis
- Generate Results

June 16

Both Mac Users

- Export Processed Results
- Send Results to Windows Team

Presentation Responsibilities

Mac User 1

- Explain MongoDB
- Explain Data Ingestion

Mac User 2

- Explain PySpark
- Explain Analysis Results

Success Criteria

✓ Dataset successfully loaded into MongoDB

✓ PySpark successfully connected to MongoDB

✓ At least 3 analyses completed

✓ Processed results exported

✓ Results provided to Windows team for Power BI visualization